MYMART CHARGEBACK DETECTION — EXECUTIVE BRIEF

Prepared for the CFO | WID3011 Deep Learning | Student ID: s2106194 | Confidential

A PROBLEM

MyMart processes e-commerce transactions across Malaysia, serving customers through a fast-moving online retail platform. Although most orders are legitimate, approximately **3.0%** of all transactions result in chargebacks, where the bank reverses the payment after the goods have already been dispatched. Each chargeback costs the company **RM 85**, which includes the value of the lost goods, payment reversal fees, and the administrative resources needed to handle the dispute. The fraud detection model supports a real-time business decision: whether a transaction should proceed normally or be held for further review before fulfilment. Missing a genuine chargeback can lead to a direct loss of **RM 85** per case, while incorrectly flagging a legitimate transaction may not have the same immediate financial impact but can frustrate customers, damage trust, and increase operational workload. For this reason, the CFO must carefully balance these two types of errors when determining the most appropriate approval threshold for the model.

B DATA

The dataset consists of **42,000** MyMart transactions and includes **12 attributes**, covering information such as transaction date, customer ID, Malaysian state, product category, order value (MYR), item count, payment method (including e-Wallet, BNPL, and Credit/Debit Card), new-customer status, device type, account age in days, and the chargeback outcome. During data preparation, missing values were identified in two features: state, which contained **420 null values (1%)**, and account_age_days, which contained **1,680 null values (4%)**. These missing values were handled by replacing unknown states with **"Unknown"** and imputing account age using the median value. A key challenge of this dataset is the highly imbalanced target variable, as only **3.0%** of **transactions are chargebacks (1,262 out of 42,000)**. Because fraudulent transactions are so rare, standard accuracy can give a misleading impression of model performance, making it necessary to apply specialised techniques designed for imbalanced classification problems. The dataset also captures several characteristics of the Malaysian e-commerce environment, including differences in purchasing behaviour across states, the growing adoption of Buy Now, Pay Later (BNPL**)** services, and seasonal shopping surges during major events such as Hari Raya and the 11.11 sales campaign. These factors create underlying patterns that the model must learn in order to effectively distinguish between legitimate and potentially fraudulent transactions.

C MODEL

The selected model architecture is a Multi-Layer Perceptron (MLP) consisting of two hidden layers with **128** and **64 neurons**, enhanced with Batch Normalisation, LeakyReLU activation, and Dropout regularisation. This architecture was chosen because fraud within the dataset is rarely driven by a single factor. Instead, chargebacks often arise from complex interactions between features such as order value, device type, payment method, and account age. While traditional models such as decision trees tend to evaluate these factors separately, the hidden layers of an MLP are better suited to learning these non-linear relationships and identifying subtle fraud patterns.

To address the significant **97%/3% class imbalance**, the training pipeline combines SMOTE oversampling, which generates synthetic examples of the minority class, with a class-weighted loss function that penalises chargeback misclassifications approximately **16 times more heavily** than mistakes involving legitimate transactions. This encourages the model to pay greater attention to fraud cases rather than simply optimising for overall accuracy.

To further improve reliability, Early Stopping and a ReduceLROnPlateau learning rate scheduler were implemented during training. These techniques help prevent the model from overfitting the synthetic samples generated by SMOTE and ensure more stable learning throughout the training process. Overall, this architecture was selected as the preferred solution because it provides a strong balance between predictive capability and training stability, making it well suited for a fraud detection problem involving large-scale tabular transaction data.

D EVALUATION

3%	90%	0.61	0.06
Precision	Recall	AUC-ROC	F1-Score

What do these numbers mean for MyMart:

The model successfully identified 227 out of 252 chargeback transactions in the held-out test set, resulting in a 90% recall rate. In practical terms, this means the system can detect approximately 9 out of every 10 fraudulent orders before they are fulfilled, significantly reducing potential financial losses. However, this strong fraud detection performance comes with a trade-off. The model's 3% precision indicates that only 3 out of every 100 transactions flagged as suspicious are chargebacks, while the remaining 97 transactions are legitimate orders that would be unnecessarily sent for review or delayed.

This high number of false alarms is a direct consequence of prioritising fraud detection by using a relatively low decision threshold of 0.28. By lowering the threshold, the model becomes more aggressive in identifying potential fraud, increasing the likelihood of catching chargebacks but also increasing the number of legitimate transactions that are incorrectly flagged.

The model achieved an AUC-ROC score of 0.61, which suggests it performs modestly better than random guessing (0.50) when ranking transactions according to fraud risk. While this may appear relatively low, it reflects the inherent difficulty of the problem, particularly given that fraud represents only 3% of all transactions in the dataset.

Importantly, these results highlight why accuracy alone is not an appropriate measure of success. A naive model that predicts every transaction as legitimate would achieve approximately 97% accuracy, yet it would fail to detect any chargebacks and therefore recover RM 0 in losses. In contrast, the current model sacrifices precision to identify most fraudulent transactions, allowing MyMart to prevent a substantial portion of potential chargeback losses. The final decision depends on whether the financial benefits of preventing fraud outweigh the operational costs associated with reviewing a larger number of flagged transactions.

E DEPLOYMENT CHALLENGES

#	Challenge	Why It Matters for MyMart	Recommended Mitigation
1	**Model Drift**	Fraud patterns do not remain constant throughout the year. During major shopping periods such as Hari Raya, 11.11, and 12.12, transaction volumes increase significantly and fraudsters often adopt new tactics. As a result, a model trained on historical data may gradually become less accurate over time.	Retrain the model every quarter using the most recent 6 months of transaction data. In addition, implement an automated monitoring rule that triggers an early retraining cycle whenever the weekly chargeback rate differs by more than **15%** from the previous 4-week average.
2	**Prediction Latency**	The fraud detection system must generate a risk score within **200ms** while handling peak traffic of up to **800 transactions per minute**. If predictions take too long, customers may experience delays during checkout, affecting the overall shopping experience.	Convert the trained model into ONNX format to improve inference speed on CPUs. Cache the scaler parameters during system startup and conduct load testing at **800 transactions per minute** before deployment to ensure performance requirements are met.
3	**Feedback Loop Bias**	When a transaction is flagged and held for review, some legitimate customers may abandon their purchase or choose a different payment method. Over time, this can bias future training data because the model is learning from transactions that have already been influenced by its own decisions.	Maintain separate records for transactions flagged by the model and those that are confirmed chargebacks. Use only verified chargeback outcomes for retraining. Additionally, allow a small random sample of flagged transactions to proceed without intervention to preserve unbiased ground-truth data.
4	**Lack of Interpretability**	MyMart's banking partners may require clear explanations for why certain transactions are classified as high risk. Since the MLP functions largely as a black-box model, it cannot directly explain its decisions, creating potential compliance and reporting challenges.	Implement SHAP explanations alongside the model. For every flagged transaction, automatically generate the top contributing factors, such as *new customer*, *BNPL payment method*, or *high order value*, and include these explanations in reports sent to banking partners.
5	**Class Imbalance Over Time**	The live transaction stream remains heavily imbalanced, with approximately **97% legitimate transactions** and only **3% chargebacks**. If future retraining is performed on raw production data without balancing techniques, the model may gradually become less effective at detecting fraud.	Apply SMOTE or class-weighted training during every retraining cycle. Establish a monitoring alert that triggers an investigation whenever the rolling **2-week r**ecall on confirmed chargebacks falls below **85%**.
6	**Data Pipeline Integrity**	The model relies on **11** real-time input features, including fields such as account_age_days and state, which contained missing values during training. Any upstream system failure, schema change, or unexpected data issue could introduce invalid inputs and lead to unreliable predictions.	Implement a validation layer before each prediction. Transactions containing excessive missing values, unexpected categories, or values outside the training distribution should be flagged or rejected. All such cases should be logged and reviewed regularly by the data engineering team.

F BUSINESS VALUE & ROI

Using the model's actual 90.1% recall and the RM 85 chargeback cost, the estimated annual net benefit under conservative assumptions is RM 61,495, assuming false positives are handled by an automated risk-alert workflow at RM 1 per flagged transaction rather than a full manual review.

Line Item	Value	Assumption / Source
Annual transaction volume	42,000	Full dataset size; assumed representative of annual throughput
Annual chargeback count	1,262	Actual count from dataset (1,262 / 42,000 = 3.0%)
Recall (actual from Section A)	90.1%	227/252 chargebacks caught on held-out test set
Chargebacks prevented per year	1,137	1,262 × 0.901
Gross savings (RM 85/chargeback)	RM 96,645	1,137 × RM 85
Precision (actual from Section A)	3%	True positives / all flagged; reflects high FP rate at 90% recall threshold
False Positives in test set	7,030	From confusion matrix: TN=1,118, FP=7,030, FN=25, TP=227
Annual false positives (FP)	35,150	7,030 × (42,000 / 8,400)
FP handling cost	RM 35,150	Assumes RM 1 per FP via automated alert; no manual hold (conservative minimum)
Annual system & infra cost	RM –	Assumed absorbed by existing data platform; quoted separately by IT
Estimated Net Benefit	**RM 61,495**	**RM 96,645 – RM 35,150**
Break-even FP cost	RM 2.75/FP	Net benefit = 0 when FP cost reaches RM 96,645 / 35,150

Note: If false positives trigger manual agent review at RM 3–5 each, net benefit turns negative (RM –8,805 to RM –79,105). CFO approval should be contingent on deploying an automated alert workflow — not a human review queue — to keep FP costs below the RM 2.75 break-even.

Ethics ETHICAL RISK & GOVERNANCE

Named Risk: Algorithmic Bias (Vieira Ch. 7)

The analysis identified is_new_customer and payment_method as the two most influential features in the model, as confirmed by the permutation importance results presented in Section A. Within the Malaysian e-commerce landscape, new customers and users of BNPL (Buy Now Pay Later) services are more commonly associated with younger and lower-income demographic groups. This creates a potential risk that the model may unintentionally treat these customers differently.

At the selected operating point, the model achieves a **90% recall rate**, successfully detecting most fraudulent transactions. However, this comes at the cost of a very low **precision of 3%**, meaning that only **3 out of every 100 flagged transactions** are actually fraudulent, while the remaining **97 transactions are legitimate orders**. The confusion matrix further highlights this issue, showing **7,030 false positives** compared to only **227 true positives** within the **8,400-sample test set**. If a large proportion of these false positives are concentrated among new customers or BNPL users—a realistic possibility given the strong influence of these features—the model could create unintended disadvantages for specific customer groups. These customers may experience additional verification checks, delayed order fulfilment, declined purchases, or a generally poorer shopping experience, ultimately reducing trust in the platform.

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Governance Recommendation: Before deploying the model into production, MyMart should conduct a Disparate Impact Audit to examine whether false positive rates differ significantly across customer groups. In particular, the audit should analyse results separately for **(a) the is_new_customer flag** and **(b) payment_method categories**. If the false positive rate for new BNPL customers exceeds the overall false positive rate by more than **20 percentage points**, corrective action should be taken. This may involve increasing the decision threshold for that customer segment or developing a separate recalibrated model tailored to that group. To ensure ongoing fairness and accountability, the audit should be repeated on a quarterly basis, with summary findings reported to MyMart's Risk Committee for review and oversight.

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*All metrics referenced in this discussion are taken directly from the Section A notebook outputs (**Cell 179** and **Cell 182**): **Precision = 0.03**, **Recall = 0.90**, **AUC-ROC = 0.61**, **test set size = 8,400 samples**, and the confusion matrix values of **TN = 1,118**, **FP = 7,030**, **FN = 25**, and **TP = 227**.*